NUST Institute of Civil Engineering (NICE-SCEE)
National University of Sciences & Technology (NUST)

CS114: Fundamentals of Programming

Class: BECE2018

Lab 08: Control Statements – II

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Lab 08: Control Statements – II

In this lab you will practice using **while and for loops** to repeat a block of code. First you will review some programs that demonstrates the use of loops, then you will write your own C++ code to practice these skill.

Our aims today are to practice on:

- while loop
- for loop

The syntax template for a **for** statement is

```
while(condition)
{
      statement(s);
}
```

The syntax template for a **for** statement is

```
for(initialization; test; update)
{
      statement(s);
}
```

Refer to the lecture material shared on LMS for help.

Example 1

```
#include <iostream>
using namespace std;

int main()
{
    //for loop execution
    for (int a=10; a<20; a=a+1)
    {
        cout << "Value of a: " << a << endl;
    }

    return 0;

}
```

Example 2

```
#include <iostream>
using namespace std;

int main()
{
    //a rectangle comprising of x's
    for (int row=0; row<4; row++)
    {
        for (int col=0; col<12; col++)
        {
            cout << "x";
        }

        cout << "\n";
    }
    return 0;
}
```

Example 3

```
#include <iostream>
using namespace std;

int main()
{
  int n, i;

  cout << "Enter a positive integer: ";
  cin >> n;

  for(i = 2; i <= n / 2; ++i)
  {
    if(n % i == 0)
    {
        cout << n << " is not a prime number as it is divisible by " << i << endl;
        break;
    }
    else {
        cout << n << " is a prime number" << endl;
        break;
    }
  }

  return 0;
}
```

Example 4

```
#include <iostream>
using namespace std;

int main()
{
  int i = 0;

  //skipping initialization statement in for loop
  for(; i <= 10 ; i++)
  {
        cout << "\n" << i;
  }

  return 0;
}
```

Deliverables

1. Execute all **Tasks** given in this document and take screen shots of the code and outputs of each task.

Upload single word document on LMS. Follow the naming convention for file. No late submissions will be accepted.

Deadline: Check deadline of the submission link on LMS.

Task 1:

Two numbers x and y are entered through the keyboard. Write a program to compute the value of x raised to the power of y using while loop.

Task 2:

Write a program that takes a number from user and computes its factorial using while loop.

Task 3:

Write a program to calculate and display the **sum** and **average** of 1 to **n** in steps of **s**, where **n** and **s** are integer-type inputs from the keyboard such that **s** denotes the step size between 1 and **n**. For example, if n=21 and s=3 are passed, it will display the sum and average of 1 + 4 + 7 + 10 + 13 + 16 + 19.

Task 4:

Write a program that prints asterisk sign (*) in a way shown below using for loops

```
**************
*************
************
***********
**********
*********
********
*******
******
*****
****
***
**
*
```

Task 5:

Write a program that takes a number from user and computes its factorial using ***for*** loop.

Task 6:

Write a program that takes a number from user and displays its multiplication table on screen using for loop.

Example: if a user gives 2 as input, output should be the following:

$2 \times 1 = 2$
$2 \times 2 = 4$
$2 \times 3 = 6$
$2 \times 4 = 8$
$2 \times 5 = 10$
$2 \times 6 = 12$
$2 \times 7 = 14$
$2 \times 8 = 16$
$2 \times 9 = 18$
$2 \times 10 = 20$